

Here we outline a step-by-step guide to installing OpenStack with Tacker, an orchestrator designed to simplify the deployment and management of VNFs and enable complex network topologies. We have divided this work into two parts: this first part covers the installation of OpenStack with Tacker, while the second part next month will focus on VNF deployment on OpenStack, and Tacker's crucial role in managing NFV and SDN technologies.

Benefits of OpenStack

There are several benefits of using OpenStack as a platform for building and managing cloud computing infrastructure.

- OpenStack is highly scalable, making it easy for users to add and remove resources as required.
- It supports a variety of use cases, including bare metal servers, containers, and traditional virtual machines. Additionally, it is highly adaptable, allowing customers to customise the platform to meet their unique requirements.
- It is designed to work with different software programs and hardware, making it simple to integrate with existing infrastructure.
- Since OpenStack is open source software, users can utilise the platform and all its components without any cost. This can result in significant cost savings when compared to private cloud platforms.
- It has a large and active community of developers and users, providing easy access to help and resources.
- It is vendor-neutral, independent of any one provider of hardware or software, giving users more options and flexibility when creating and maintaining their cloud infrastructure.

OpenStack components

The components of OpenStack are crucial to the platform's functionality.

Nova: This component schedules and manages computing resources such as virtual machines and containers. It is a prerequisite for any OpenStack implementation as it provides the foundation for establishing and monitoring cloud computing infrastructure.

Neutron: This service manages networking resources such as virtual networks, load balancers, and firewall rules. It is a software-defined networking project responsible for providing IP addresses to virtual machines, subnets, topologies, and traffic routing. Neutron manages all networking facets for the virtual networking infrastructure (VNI) and the access layer aspects of the physical networking infrastructure (PNI) in the OpenStack environment.

Glance: This component manages and distributes virtual machine images. Users can manage the metadata associated with their images, and store and retrieve them using Glance.

Keystone: This component is responsible for handling user authentication and permission. It offers a mechanism for managing access tokens, which are needed to authenticate users when they access the OpenStack deployment, as well as a central directory of users and the resources they are permitted to access.

Cinder: This component handles block storage devices in OpenStack. It

provides APIs to users that manage and consume block storage on their virtual instances, offering volumes to Nova virtual machines, bare metal hosts, containers, and more while ensuring high availability, fault tolerance, recoverability, and open standards.

Image: This service is provided by the Glance project, and allows users to upload and discover data assets such as images and metadata definitions that are meant to be used with other services. Glance manages virtual machines' disk images, provides image delivery to virtual machines, and provides snapshot (backup) services.

Swift: This project provides object-level storage via a RESTful API. Swift is a highly available, distributed, eventually consistent object/blob store. Organisations can use Swift to store lots of data efficiently, safely, and cheaply.

Dashboard: This project provides administrators with a graphical user interface to administer OpenStack and its various components. The horizon is the canonical implementation of OpenStack's dashboard, which provides a web-based user interface to OpenStack services, including Nova, Swift, Keystone, and more.

OpenStack architecture

The OpenStack architecture is designed in a way that the compute, network, and storage components are exposed through

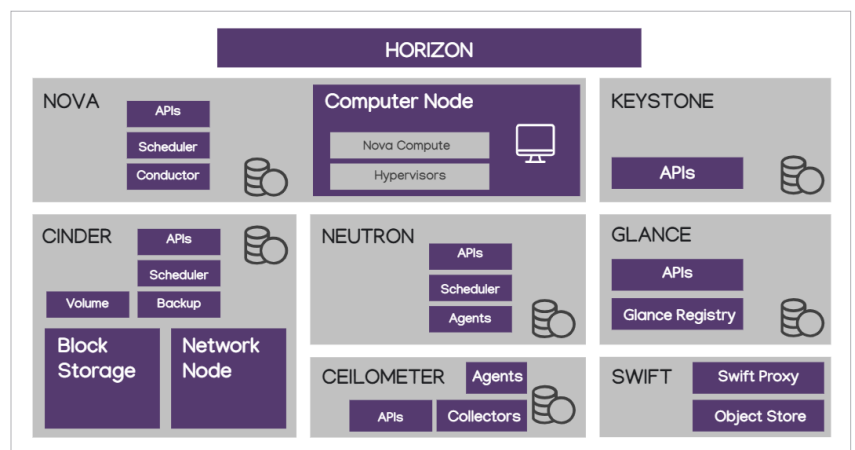


Figure 1: Overview of OpenStack architecture